vocabulary and key patterns, that increases from lower to upper layers (Section 4). Is this because the layer's output space transforms across layers? If so, how? We note that this possible transformation cannot be explained solely by the function of feed-forward layers: if the model only did a series of key-value look-ups and value-distribution aggregation via weighted addition, then a single, unifying embedding space would appear more natural. Thus, the transformation might have to do with the interplay between feed-forward layers and self-attention layers.

- **Beyond language modeling.** Our formulation of feed-forward networks as key-value memories generalizes to any transformer model, e.g. BERT encoders and neural translation models. We thus expect our qualitative empirical observations to hold across diverse settings, and leave verification of this for future work.

- **Practical implications.** A better understanding of feed-forward layers has many implications in NLP. For example, future studies may offer interpretability methods by automating the pattern-identification process; memory cells might affect training-data privacy as they could facilitate white-box membership inference (Nasr et al., 2019); and studying cases where a correct pattern is identified but then suppressed during aggregation may guide architectural novelties.

Thus, by illuminating the role of feed-forward layers, we move towards a better understanding of the inner workings of transformers, and open new research threads on modern NLP models.

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A Pattern Analysis

Table 3 provides a fully-annotated example of 25 prefixes from the memory cell $\mathbf{k}^5_{895}$.

B Implementation details

In this section, we provide further implementation details for reproducibility of our experiments.

For all our experiments, we used the language model of Baevski and Auli (2019) (247M parameters) trained on WikiText-103 (Merity et al., 2017). Specifically, we used the model `transformer_lm.wiki103.adaptive` trained with the fairseq toolkit[6].

WikiText-103[7] is a well known language modeling dataset and a collection of over 100M tokens extracted from Wikipedia. We used spaCy[8] to split examples into sentences (Section 3).

[6] https://github.com/pytorch/fairseq

[7] https://blog.einstein.ai/the-wikitext-long-term-dependency-language-modeling-dataset/

[8] https://spacy.io/

1	It requires players to press
1	The video begins at a press
1	The first player would press
1	Ivy, disguised as her former self, interrupts a Wayne Enterprises press
1	The video then cuts back to the press
1	The player is able to press
	Leto switched
1	In the Nintendo DS version, the player can choose to press
1	In-house engineer Nick Robbins said Shields made it clear from the outset that he (Robbins) "was just there to press
1	She decides not to press
1	she decides not to press
1	Originally Watson signaled electronically, but show staff requested that it press
1	At post-game press
1	In the buildup to the game, the press
2	Hard to go back to the game after that news
1	In post-trailer interviews, Bungie staff members told gaming press
	Space Gun was well received by the video game
1	As Bong Load struggled to press
	At Michigan, Clancy started as a quarterback, switched
1	Crush used his size advantage to perform a Gorilla press
1,2	Groening told the press
1	Creative director Gregoire <unk> argued that existing dance games were merely instructing players to press
1,2	Mattingly would be named most outstanding player that year by the press
1	At the post-match press
1,2	The company receives bad press

ID	Description	shallow / semantic
1	Ends with the word "press"	shallow
2	Press/news related	semantic

Table 3: A pattern annotation of trigger examples for the cell memory $\mathbf{k}^5_{895}$. Trigger examples are annotated with repetitive patterns (upper table), which are classified as "shallow" or "semantic" (bottom table).